

Mohammad-Jamiu Babatunde Balogun

Kano, Nigeria | balogunmohammedjamiu@gmail.com | +234 708 046 2912 | [ORCID](#) | [GitHub](#) | [LinkedIn](#)

EDUCATION

Bayero University, Kano (BUK)	December 2017 - May 2024
B.Eng. Telecommunications Engineering, CGPA 4.64/5.00 (First Class Honors), Best Graduating Student	Kano, Nigeria
• Thesis: GSM-Based Weather Reporting System for Base Transceiver Station Using Arduino UNO and GSM SIM900. • Supervisor: Dr. Nasiru Bello Kadandani. • Scholarships & program awards: Bayero University Scholarship Award (Level 300); Bayero University Scholarship Award (Level 400); Best Graduating Student Award, Telecommunications Engineering. • Selected coursework: Computer Networks; Wireless Communications; Digital Signal Processing; Data Communications; Telecommunication Systems; Embedded Systems; Probability and Statistics; Programming; Engineering Mathematics.	

RESEARCH INTERESTS

Computer Networks and Network Security; Cybersecurity; AI Security; Secure Software Systems; Software Vulnerability Analysis; Large Language Model Security Evaluation; Secure Code Generation; Applied Machine Learning.

PUBLICATIONS

• Balogun, M. B.*, Geza, A. S., & Jimada, Y. U. (2026). Security Benchmarking of AI-Generated Web Application Code: An OWASP-Based Comparative Study of Coding LLMs. World Journal of Advanced Engineering Technology and Sciences, 19(3), 009-017. https://doi.org/10.30574/wjaets.2026.19.3.0292 • Umar, A. A., Salisu, A. N., Ajidoku, L., Balogun, M. B., & Abdulhadi, Z. A. (2026). An IoT-Enabled High-Frequency Smart Energy Meter with Web-Based Data Management and Control. World Journal of Advanced Engineering Technology and Sciences, 18(1). https://doi.org/10.30574/wjaets.2026.18.1.0041
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RESEARCH EXPERIENCE

AI Security Benchmarking Project	2025 - 2026
Independent Researcher	Kano, Nigeria
• Conceived and executed a comparative security assessment of AI-generated web application code. • Developed an evaluation framework based on OWASP Top 10 security risks. • Conducted static application security testing using Semgrep OSS. • Performed manual validation of security findings and vulnerability classification. • Produced reproducible benchmarking methodology suitable for future AI security studies.	
Department of Telecommunications Engineering, Bayero University	2023 - 2024
Undergraduate Researcher	Kano, Nigeria
• Designed and implemented a GSM-enabled weather monitoring system for Base Transceiver Stations. • Developed sensor acquisition and GSM communication modules using Arduino UNO and SIM900. • Collected and analyzed environmental data affecting BTS performance. • Built a low-cost prototype for remote monitoring in resource-constrained environments.	

OPEN SOURCE SOFTWARE

• Cite Formatter (Creator and Maintainer, 2026–Present): TypeScript open source package. Citation compilation engine supporting APA, IEEE, Chicago, MLA, Harvard, and Vancouver; continuous numbering for multi-chapter documents; custom formatting strategies, BibTeX export, CLI, automated testing, and CI. Built for AI-assisted academic writing. Technologies: TypeScript, Node.js, GitHub Actions, NPM. https://www.npmjs.com/package/cite-formatter • PDF Context Extractor (Creator and Maintainer, 2026–Present): Local-first research productivity application. Browser-based PDF extraction with page selection, text extraction, search, export, and AI-ready prompts; privacy-preserving (documents never leave the device). Technologies: React, TypeScript, Vite, Zustand, PDF.js. https://github.com/Mohammadjamiu/PDF-Context-Extractor
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TECHNICAL PROJECTS

MonoEd Africa	2025 - Present
Founder and Lead Developer	Nigeria
- Founded an educational technology platform providing AI-powered academic and career tools for students. - Developed tools supporting SIWES documentation, project proposal generation, research discovery, citation verification, CV generation, and academic productivity. - Designed and implemented platform architecture, product strategy, and technical development. - Built solutions used by students across Nigerian universities and polytechnics.	

INDUSTRIAL EXPERIENCE

M. Yahaya and Associates	December 2022 - May 2023
Telecommunication Engineering Intern	Kano, Nigeria
- Conducted GIS-based analysis using ArcGIS JavaScript tools. - Diagnosed and resolved networking and hardware issues. - Processed and analyzed engineering datasets using Microsoft Excel. - Supported technical field operations and infrastructure maintenance.	
M. Yahaya and Associates	October 2021 - December 2021
IT Support Intern	Kano, Nigeria
- Troubleshooted routers, computer systems, and network equipment. - Produced technical maintenance documentation. - Assisted with survey data management and reporting.	

TEACHING AND MENTORING

Bayero University	2019 - 2024
Academic Tutor and Student Mentor	Kano, Nigeria
- Mentored more than 150 engineering students. - Facilitated workshops in Python programming, MATLAB, AutoCAD, and embedded systems. - Assisted students with academic projects and technical problem solving.	
NUESA Elite Engineering Fellowship	January 2024
Team Lead	Kano, Nigeria
- Led practical workshops on Python programming and software development tools. - Guided engineering students in building introductory software projects. - Supported technical skill development beyond classroom instruction.	

LEADERSHIP

TechPlaybook	2026 - Present
Founder	Nigeria
- Co-founded an engineering and AI readiness initiative focused on practical technical skills development. - Coordinate learning programs in software engineering, AI, embedded systems, and innovation.	
Muslim Corpers Association of Nigeria (MCAN), Kano	2025
Secretary General	Kano, Nigeria
- Coordinated organizational projects and facility improvement initiatives. - Developed digital tools to improve communication and member engagement.	

CERTIFICATIONS

- Cisco Junior Cybersecurity Analyst (2024)
- Cisco Network Support and Security (2024)

PROFESSIONAL MEMBERSHIPS

- Graduate Member, Nigerian Society of Engineers (NSE)

- Member, Nigerian Institute of Electrical and Electronics Engineers (NIEEE)
- Member, International Association of Engineers (IAENG)

AWARDS AND HONORS

- Best Graduating Student, Telecommunications Engineering, Bayero University
- Bayero University Scholarship Award (Level 300)
- Bayero University Scholarship Award (Level 400)
- Award of Excellence, Nigerian Universities Engineering Students Association (NUESA)

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, TypeScript, C/C++, SQL
- **Cybersecurity:** OWASP Top 10, Semgrep OSS, Static Application Security Testing (SAST), Secure Coding Principles
- **Networking:** Computer Networks, Network Troubleshooting, Routing and Switching Fundamentals, Network Security Concepts
- **Data analysis:** Python, Pandas, NumPy, Statistical Analysis, Data Visualization
- **Software development:** React, Node.js, Git, GitHub Actions, REST APIs
- **Embedded systems:** Arduino, Proteus, Fritzing, GSM Systems

LANGUAGES

- English (Fluent)
- Yoruba (Native)
- Hausa (Intermediate)
- Arabic (Basic)

REFERENCES

Available on request.